

# BISWARUP MUKHERJEE

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## EDUCATION

**M.Sc. in AI for Computer Vision (IPC Vai)** | Bordeaux, France

**Sep 2025 – Jul 2027**

*Université de Bordeaux (UBx) - Erasmus Mundus Scholarship*

*GPA: 4.8 / 5*

- **Coursework:** Deep Learning for Visual Processing, Multimodal Sensor Fusion, High Performance Computing

**B.Tech in Computer Science and Engineering** | Tamil Nadu, India

**July 2019 – Aug 2023**

*Vellore Institute of Technology (VIT)*

*GPA: 9.2/10*

## EXPERIENCE

**Vaelsys España - Intel Solutions**

**June 2026 – Sept 2026**

**Computer Vision Engineer - Intern**

Madrid, Spain

- Deployed CCTV based action recognition & video-pose estimation pipeline for Mercadona retail-security team, using finetuned Qwen 3 VL-embeddings. Achieved **3.6× inference speedup**; Improved action ranking & clip-level anomaly AUC **from 0.69 to 0.86**.
- Built an edge CCTV-based threat-detection pipeline for EV charging stations deployed on Intel Core Ultra chips. Benchmarked INT 8 quantized VLMs (Qwen3-VL, MiniCPM-V, Intern-Video2) and ML classifiers (SVM, XGBoost) under group-aware cross-validation. Achieved **96% theft recall**.
- Optimized multi-camera MiniCPM-V inference through OpenVINO Model Server & gRPC with temporal window de-dup, and concurrent camera scheduling. Calibrated INT8 quantization and Sustained **8–12 parallel camera streams at 30 classifications/min** on a 2-stage NPU-GPU architecture [[MiniCPM PR](#)].

**PepsiCo GBS**

**Aug 2023 – Aug 2025**

**Data Science - Predictive Analyst**

Hyderabad, India

- Developed multi-stage modelling framework integrating foundation models (**Chronos-Bolt, TimesFM 2.0**) with external covariates (PnL signals), for global **Operating Cash Flow (OCF)** forecasting, resulting in **21% reduction** in Mean Absolute Error(MAE).
- Reduced computational **costs by 46%**, optimizing distributed MLOps pipelines using efficient parallel stage jobs orchestration and resource allocation in **Azure Databricks**, contributing to **\$ 1M+** in savings. [[Contributions](#)]
- Built a **GPU-accelerated** training and inference pipeline using NVIDIA RAPIDS (cuML), achieving **1.5× speedup**, ensuring numerical stability and accuracy consistency across high-impact revenue signals.

## SKILLS

**Programming & Software::** Python, C++, SQL, Git, Linux

**Machine Learning & Experimentation::** PyTorch, TensorFlow, Scikit-learn, SVM, XGBoost, Prophet, PySpark.

**NLP & Foundation Models::** Hugging Face Transformers, Qwen3-VL, MiniCPM-V, InternVideo-2, Chronos-Bolt, TimesFM 2.0, RAG, Part-Aware-Transformer for Re-ID, Gaussian Splatting

**MLOps, Deployment & HPC::** MLflow, CUDA, Azure Databricks (GPU Clusters), AWS (SageMaker, Lambda, EC2), OpenVINO (gRPC), Optimum CLI, NVIDIA RAPIDS (cuML), INT8/INT4 Quantization

## PROJECTS

**GPU-Accelerated DBSCAN Pre-Association for Scalable Multi-Object Matching** [[Github](#)] | *Python* | *RAPIDS*

- Built a GPU-accelerated DBSCAN pre-association stage before Hungarian assignment, reducing the effective  $O(N^3)$  matching workload in dense scenes.
- Benchmarked 500–10,000 detections, identifying a crossover at  $\sim 4,000$  detections and achieving up to **5.8× speedup** at 10k scale.

**Onboarding Assistant - Multi AI Agent** [[Github](#)] | *Python* | *Bedrock Multi-Agent RAG* | *Lambda* | *SQL*

- Designed a multi-agent RAG system using Amazon Bedrock with embedding-based retrieval and LLM reasoning; optimized top-k retrieval and prompt composition to balance answer accuracy and latency.
- Deployed a cloud-native architecture using EC2, SageMaker, and AWS Lambda for scalable inference, agent orchestration, and serverless API integration.

## ACHIEVEMENTS

- Full European Commission Scholarship (€50,000+) awarded to 17 out of 1600+ global applicants for Masters.
- Winner(3rd place): CISPA AI & Cybersecurity Hackathon 2026: Attacking LLM watermarking and Model privacy.
- Won the AI CLIMATE X Hackathon in sustainable AI and potential incubation is in progress by IncubatorZero.